

AMD BC-250

How to

The ASRock AMD BC-250 is a crypto mining hardware. It's not a card but a full computer based on the PlayStation 5. The BC-250 APU is slightly tuned down to 6 CPU cores (PS5 has 8) and 24 GPU compute units (PS5 has 32). The memory is 16GB integrated GDDR6.

IMPORTANT!

The following description includes hardware modifications involving sharp tools and exposure to 110 V electrical components, as well as software modifications that can render the BC-250 device non-functional. Proceed only if you fully understand the risks and have the necessary skills and safety knowledge.

By using this information, you acknowledge that you are solely responsible for your actions. I do not accept any liability for injuries, electrical hazards, data loss, device damage, or any other consequences resulting from following the procedures in this document. If you are not completely confident in performing these modifications safely, do not attempt them.

Hardware

Case

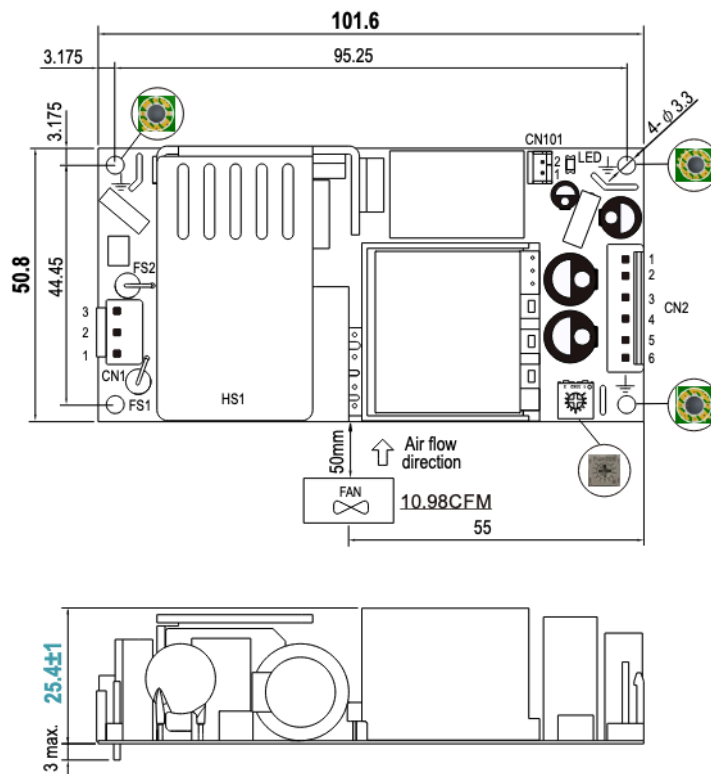
I found 2 nice 3D designs for the BC-250. Both offer integrated PSU and cooling.

Mini case with integrated LOP 300-12 PSU

Probably the most compact shell case with integrated PSU. The PSU that matches the design is the LOP 300-12 (300W 12V). It does not come with cooling, one central 120mm fan provides the cooling both for the PSU and the BC-250 board. This looks ideal to make it a travel PC. No modification needed for the finstack.



To use the LOP 300-12V power supply you have to remove the factory pinout and solder a PCIe power connector instead. Same thing with the 110V power plug.



AC Input Connector (CN1) : JST B3P-VH or equivalent

Pin No.	Assignment	Mating Housing	Terminal
1	AC/L	JST VHR or equivalent	JST SVH-21T-P1.1 or equivalent
2	No Pin		
3	AC/N		

DC Output Connector (CN2) : JST B6P-VH or equivalent

Pin No.	Assignment	Mating Housing	Terminal
1,2,3	+V	JST VHR or equivalent	JST SVH-21T-P1.1 or equivalent
4,5,6	-V		



I've found that the middle shell piece was very tight so I've modified the original model. This new version is easier to install. The design uses M3 size heated inserts to assemble the case and hold the parts. The 3D model files are included in the repo with the original author's instruction.

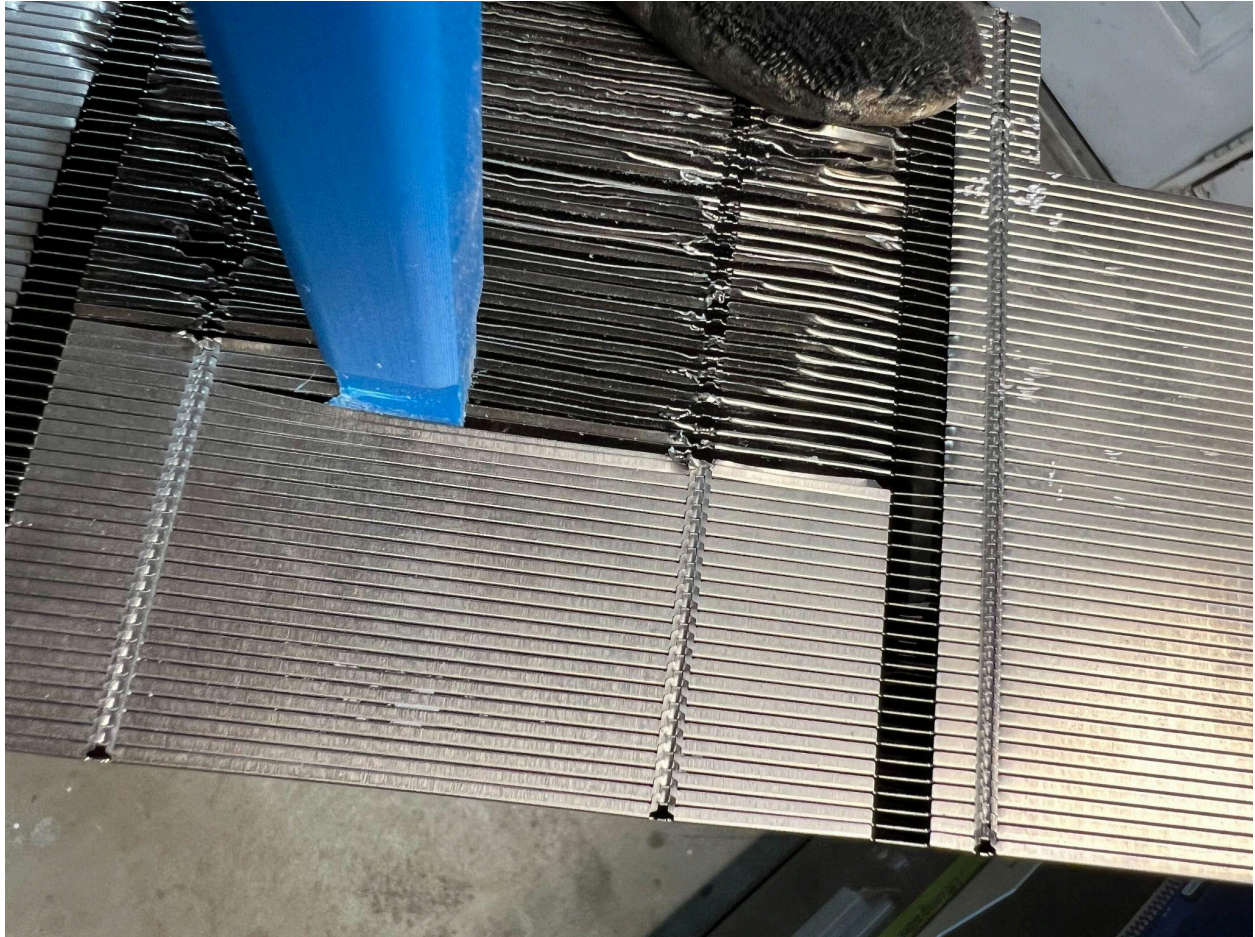
Credit:

<https://www.printables.com/model/1228207-asrock-amd-bc-250-shell-case/files>

Case with Flex PSU

Good size case. The cooling performance is probably better than with the mini shell case. The Flex PSU is probably more reliable. Finstack has to be opened up which is not easy to do. I found a finstack scooper 3D design and used that. I only opened up the middle piece for the single fan design.

The technique I've used was to get the scooper in position at the small openings at the middle section of the finstack, and then hammered it through till the other opening. That gave me a ~120x120mm open fin in the middle section as this case requires it.



The case is made up of a front and back piece and 2 custom adapters. Printing the front and back piece was over 2 days individually.



The 3D model files are included in the repo with the original author's instruction.

Credit:

<https://www.printables.com/model/1423572-minimal-case-for-bc250-and-flex-psu/files>

<https://www.printables.com/model/1282906-bc-250-scooper/files>

Cooling

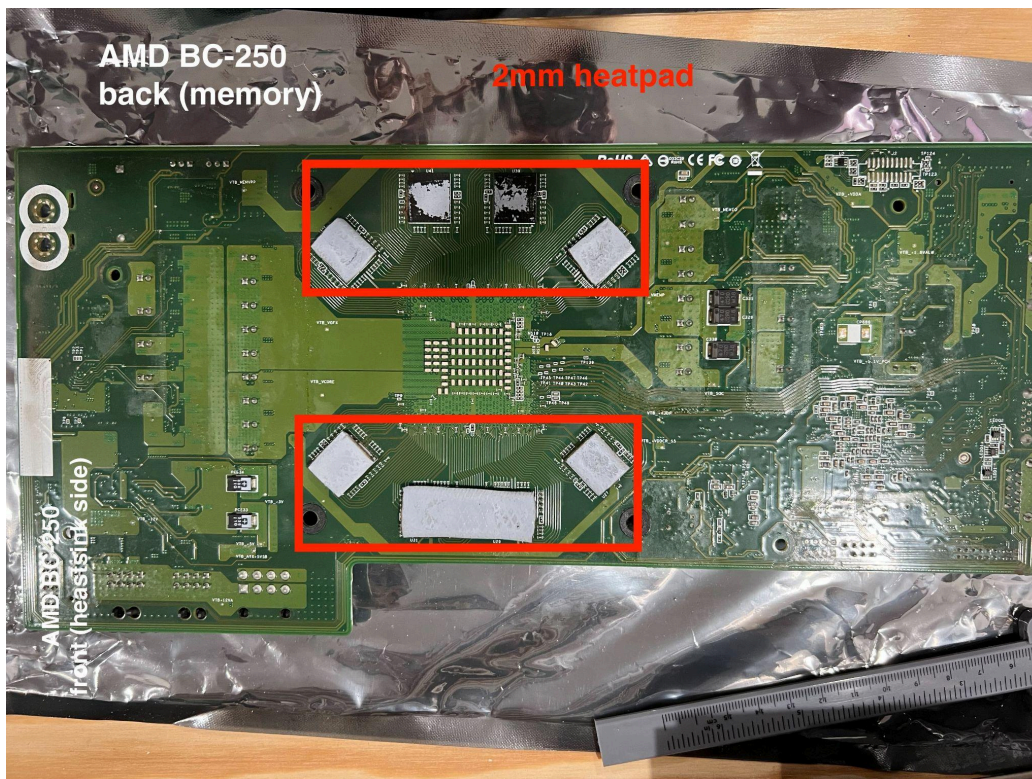
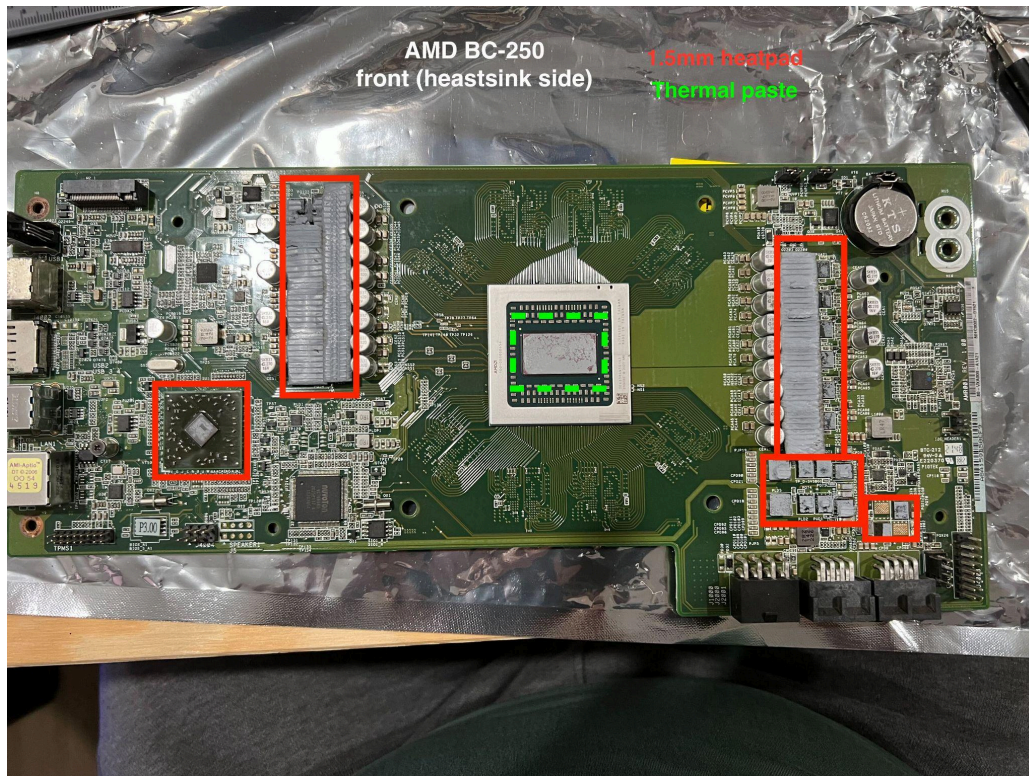
Depending on your choice of case you might have to open the closed server style finstack. Some open up completely and use 2x120mm fans. Some just open the middle section which is a 120x120 area that perfectly fits a single 120mm fan.

Open the heatsink

It's not easy to do. This scooper has helped a lot:

<https://www.printables.com/model/1282906-bc-250-scooper/files>

Headpads replacement



BIOS

In order to use this as a gaming device, the shared memory allocation has to be changed. For that you need an unlocked 3.0 Bios version.

Flash

There's no need for hardware flashing the BIOS any more. Lots of descriptions and videos from a year ago has shown that. I found a EFI Flasher tool that works with the "unlocked" 3.0 BIOS.

Based on:

- <https://github.com/kenavru/BC-250>
This is the flasher tool
- <https://gitlab.com/TuxThePenguin0/bc250-bios>
This has the custom bios to be able to change the memory allocation

I've created this "custom" bios flash, that has the unlocked 3.0 BIOS version

https://github.com/csabakecskemeti/amd_bc-250_how-to/tree/main/bios

Follow the instructions here:

https://github.com/csabakecskemeti/amd_bc-250_how-to/blob/main/bios/mem_unlock_bios_and_flasher/EFI%20update%20Bios%20Step.pdf

Additionally remove the NVME drive just to be safe.

Memory allocation for gaming

From: <https://github.com/mothenjoyer69/bc250-documentation>

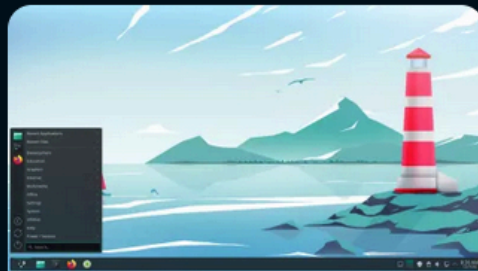
VRAM allocation is configured within: **Chipset -> GFX Configuration -> GFX Configuration**. Set **Integrated Graphics Controller** to forced, and **UMA Mode** to **UMA_SPECIFIED**, and set the VRAM limit to your desired size. 512MB is best for general APU use.

You have to do this to be able to game on the device. If this set even Cyberpunk 2077 has a playable framerate.

OS

Manjaro Linux is a good choice. It knows the hardware has drivers. I've used the KDE Plasma

<https://manjaro.org/products/download/x86>



KDE Plasma

Plasma is one of the most advanced desktops and offers great flexibility. Suitable for beginners and pros.

[Download](#)[More](#)

Utilities

Oberon-Governor

<https://gitlab.com/TuxThePenguin0/oberon-governor>

This repo has an install script

<https://github.com/pnbarbeito/bc250-arch>

Gaming

You can install Steam via the UI there's an Add/Remove Software app where you can install Steam from.

Gaming performance

Cyberpunk 2077 has playable framerate.



AI

Vulcan [manjaro linux]

```
sudo pacman -S amdvlk spirv-tools vulkan-headers vulkan-radeon  
vulkan-tools vkd3d lib32-vulkan-radeon
```

Llama.cpp for Vulkan

```
cmake -B build -DGGML_VULKAN=ON  
cmake --build build --config Release
```

Sensors

Here are the **exact shell commands** you need, plus the config file entries, formatted cleanly so you can copy/paste:

1. Load the driver immediately

```
sudo modprobe nct6683 force=true
```

 - a. Make the `force=true` option persistent
 - i. Create (or edit) this file

```
sudo nano /etc/modprobe.d/sensors.conf
```
 - ii. Add

```
options nct6683 force=true
```
 - iii. Save & exit
2. Ensure the module loads at boot
 - a. Create (or edit)

```
sudo nano /etc/modules-load.d/99-sensors.conf
```
 - b. Add

```
nct6683
```
 - c. Save & exit
3. Regenerate initramfs
For Arch-based systems

```
sudo mkinitcpio -P
```
4. After reboot, in bash run

```
sensors
```
5. Use it with `psensor`

Useful links

<https://github.com/mothenjoyer69/bc250-documentation>